A completely specified nine-positive consistency equation

Nine positive inputs; accepted evaluator: 433 operations, degree 1,628,768. All fixed data below.

The only unknowns are $b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}$. All other letters are polynomial abbreviations or fixed constants. The equation is

$$\boxed{\left[\Theta^2 + \Omega^2(a_0 - a_1)\right]^2 - 4\Omega^2 a_0 \Theta^2 = 0.}$$

Source and ratio polynomials. For any displayed finite list put $p(e_1, \dots, e_t) = \sum_{\nu=1}^{t} B^{e_\nu}$. Define

$$\begin{gathered}
B = 2^{864026} b^2, \quad N_0 = B^{117}, \quad \ell = p(\mathcal{W}), \quad Z = p(\mathcal{T}), \\
u = \ell + g, \quad G_c = D_2(B)u^2 + D_1(B)u + D_0(B), \\
S_c = g + N_0(Z + 2G_c), \\
T_c = N_0 - 1 - \ell(b-1) + N_0 Z(B-2), \\
L_c = bB^{2200}, \quad R = L_c(S_c + T_c + 1) - S_c - 1, \\
N = 2^{6480184} B^{1162}, \quad c_* = bB^{116}, \\
U = N^2 w, \quad Y = N^2 s, \quad M = RY, \\
P = 2M^2 U, \quad k = R + 1 + h(P-1), \\
A = M(U+1), \quad B_1 = 2R+1, \quad W_1 = bw, \\
J_r = 4A - 5, \quad Q_r = 3W_1 C - 2(W_1^2 - 1), \\
m = k^2(4c_* - g) - 16c_*(C - kY)^2, \quad \Omega = J_r m, \\
E_0 = J_r(n + W_1^2) + Q_r, \quad V_0 = Pk, \\
\delta = A^2 - 1, \quad D = 1 + \delta C^2, \quad e = iJ_r C^2, \quad d_e = De, \\
F_p = 1 + (\delta e) d_e, \quad \Lambda = CD + 1 - A, \quad G_p = A + F_p \Lambda, \\
H = C + F_p[B_1 + (j-1)C], \quad T_m = A d_e, \quad V_1 = T_m(G_p H), \\
I = 1 + (G_p H)^2 - H^2, \quad a_0 = 1 + V_0^2 - k^2, \quad a_1 = DF_p I, \\
\Theta = E_0 - \Omega(V_0 + V_1).
\end{gathered}$$

The complete fixed lists, with $p(\mathcal{W})$ and $p(\mathcal{T})$ meaning the sum over all entries, are

$$\begin{aligned}
\mathcal{W} = (&1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50, \\
&2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60), \\
\mathcal{T} = (&28, 124, 325, 384, 589, 630, 643, 696, 846, 1073, 1082, \\
&1159, 1251, 1261, 1450, 1510, 1627, 1756, 1793, 1807, 2028, 2050).
\end{aligned}$$

All 85 kernel terms. Equal-coefficient powers are grouped only by the explicitly defined finite-list notation p:

$$\begin{aligned}
D_0(B) &= 2K_0 B^{124} - 2p(325, 384, 1450, 1627) - 6B^{630}, \\
D_1(B) &= 2p(26, 33, 477, 542, 570, 605, 1020, 1027) \\
&\quad + 2p(1047, 1067, 1144, 1234, 1354, 1358, 1370) \\
&\quad + 2p(1501, 1691, 1719, 1733, 1756, 1918, 1941) \\
&\quad - 2p(1, 493, 662, 754, 982, 1698, 1796) - 8B^{514} - 6B^{1023} \\
&\quad - 2K_0 B^{63} - 2(K_0+1)B^{117} + 2\beta(K_0+1)B^{123} \\
&\quad - 2(M_C + M_F)B^{845} - 4\beta B^{1260}, \\
D_2(B) &= -\beta p(0, 87, 661, 843, 1795) \\
&\quad - p(29, 83, 119, 135, 338, 377, 393, 758, 812, 845) \\
&\quad - p(1076, 1176, 1256, 1295, 1512, 1632, 1794, 1807, 1843) \\
&\quad - 2p(105, 398, 611, 858, 1452, 1542) \\
&\quad + 2p(299, 326, 1326, 1521) \\
&\quad + p(384, 400, 691, 837, 1823, 1850) - 2\beta M_F B^{844} \\
&\quad - 4B^{907} - 2\beta^2 B^{1259}.
\end{aligned}$$

The complete fixed coefficient tuple. The divisions and floors below define constants only; the rational intermediate defining K_0 is an integer.

$$\begin{gathered}
\rho = 2^{89}, \quad \mathcal{B} = \rho^{4831}, \quad \beta = \mathcal{B} - 1, \\
\lambda = 2^{29}, \quad \theta = 2^{42}, \quad \mathcal{Q} = 2^{40}, \\
G_t(x) = \sum_{r=0}^{t-1} x^r, \quad \mu = 510 + 2^{17} + 2^{26} + 2^{28} + \lambda(2^{44} - 1).
\end{gathered}$$

$$\begin{aligned}
\mathcal{S} = &\{0, \dots, 456\} \cup \{495, \dots, 504\} \cup \{507, 508\} \\
&\cup \{517, \dots, 537\} \cup \{539, \dots, 549\} \\
&\cup \{567, \dots, 583\} \cup \{1203, 1807\}.
\end{aligned}$$

$$\begin{aligned}
M_C &= \mathcal{B} - 1 - \sum_{r \in \mathcal{S} \setminus \{1\}} \rho^r - 2\rho^{507}, \\
M_F &= \mathcal{B} - 1 + 4 + \mu(\rho^{2409} + \rho^{3015}) \\
&\quad + \rho^{528} G_{10}(\rho) + \rho^{585} G_{11}(\rho) + \rho^{1203} + \rho^{1807} \\
&\quad + (2^{87} - 1)\rho^{4823} G_4(\rho) + (2^{38} - 1)\rho^{4827}, \\
K_0 &= \rho^{33} + \rho^{68} + \rho^{664} + \rho^{1268} + \rho^{4877} \\
&\quad + (\rho^{2409} + \rho^{3015})\Phi(1/\rho),
\end{aligned}$$

$$\begin{aligned}
\Phi(x) = &\, O G_{457}(x) + \lambda \sum_{a=0}^{456} \ell_a x^{r_a} \\
&- (255\,2^9 + 2^{27}) x^{374} - (255\,2^{18} + 2^{27}) x^4 \\
&+ \lambda\big[(x^{539} + 2^{21} x^{517}) G_{11}(2x) \\
&\quad + (2^{11} x^{528} + 2^{32} x^{495}) G_{10}(2x) \\
&\quad + \theta(x^{539} + x^{1203} + x^{1807})\big] \\
&+ \big[(2^9 + \lambda(2 + 2^{33})) x^{567} \\
&\quad + (2^{18} + \lambda(2^{22} + 2^{33})) x^{575}\big] G_8(2x) \\
&+ (2^{27} + \lambda(2^{11} + 2^{32})) x^{583}.
\end{aligned}$$

The ranks and labels. The following 457 fixed quadruples are $t_r = (w_r, n_r, e_r, s_r)$. From the fixed integer $\mathcal{A}$ below, for $0 \le r < 430$ put

$$\begin{gathered}
b_r = \lfloor \mathcal{A} / 1024^r \rfloor \bmod 1024, \quad u_r = b_r \bmod 256, \\
v_r = \lfloor b_r / 256 \rfloor \bmod 2, \quad d_r = \lfloor b_r / 512 \rfloor, \\
t_{2+r} = ((1 - d_r)(512 + u_r), 512 + r, d_r(256 + u_r), v_r).
\end{gathered}$$

Here $a \bmod b = a - b\lfloor a/b \rfloor$. The remaining quadruples are

index	range	t_r
0		$(2, 3, 3, 1)$
1		$(5, 0, 0, 0)$
$432 + r$	$0 \le r < 2$	$(0, 942 + r, 0, 942 + r)$
$434 + r$	$0 \le r < 2$	$(0, r, 0, r)$
$436 + r$	$0 \le r < 4$	$(0, (r \bmod 2) + 942\lfloor r/2 \rfloor, 0, 6)$
440		$(2, 2, 2, 0)$
441		$(3, 4, 4, 698)$
442		$(4, 4, 4, 0)$
$443 + r$	$0 \le r < 3$	$(1, 6, 1, 2 + r)$
$446 + 3r + t$	$0 \le r, t < 3$	$(r^2, 5, t, 5)$
455		$(256, 0, 0, 512)$
456		$(0, 0, 512, 512)$

For these bound indices define

$$\begin{aligned}
&\kappa_r = 2^{30} w_r + 2^{20} n_r + 2^{10} e_r + s_r, \\
&\ell_r = 2w_r + 2^{11} n_r + 2^{22} e_r + 2^{32} s_r, \\
&r_0 = 0, \quad r_1 = 1, \\
&r_t = 2 + \sum_{r=2}^{456} \left\lfloor \frac{\mathcal{Q} + \kappa_t - \kappa_r - 1}{\mathcal{Q}} \right\rfloor \quad (2 \le t \le 456), \\
&O = 1 + 255\,2^9 + 255\,2^{18} + 2^{27} + \lambda.
\end{aligned}$$

Subscripts in the constant recipes are bound indices, not extra unknowns.

The integer $\mathcal{A}$. Digit values are $0, \dots, 61$ in the alphabet `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`

For the following 722 characters put $\mathcal{A} = \sum_{r=0}^{721} \epsilon_r 62^{721-r}$, where ϵ_r is the value of character r, starting at zero:

```
RCtYkANMQ1gmPqxuKhLmpi1mjCvu243YjyAxv9hbC2UsurRLykLFhuHVBGGUZ5
whVShVOFQcVVQSR2GS6JAW1DGYxGKlA4rJavFCW5i63LLSs8xEGvWTbLjSFULV
ODe48Kw2ghEaa5Q9PZZVHtki7C45jfVYROOzvVT9DAGjLOBlBSHQU8WPUF3SpC
kTfDQF7mUboMEAeyPWARCyClpnxtqgsit318hQgiIdKauxc6memnYNtycBOs7z
K8dMgYaLRxNjlcLvJzZ9Z5wdCG3wjd5P6ZVqC2SXVaimqXrTY5dNxbCH2B5VKA
Sp16kht0JV65p4VclYsI6zi6nIr7elpUCOX7IHdu6qKiu6Kt5MqlYZjChEzG6y
W1WsPdMB71PL3oOCFqFncCtVEMUounam1fyhduQYoTyTCIEMqcD2BPQfMVaBa9
BZHQ8k7rpS3eiM7Rsak5uUsoP1Tf1uhWshndLuN8yz7UdD5QXM6AJ3ovPdWsqO
HGBeRaR8kch4R9JVbkAxS2ByaeE2O6czIT0sWtpasWDdZB9FuDQUFYhSvPPPr8
nSQmnMkF9OCBzWkpeP81azGOWtiT2XTxpaLcR4ecaK1uZyvBleYknk6QTwEC15
XJ3nJBoP4sMTGaV6unoGzq9s1KN4KtXrVRuTpi6EuqizW9njcp9Uikq1VNjVtw
W1AGLJu3eaWgOEEF7vROqLuNbzEheYnyi4SORWVp
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Absence of positive integer solutions is equivalent to Con(ZFC) by the inherited source/native calibration and the ordinary saturated Pell argument. This page specifies the equation; it does not supply an end-to-end formal Lean proof.